

YUVARAJ AKKOLE

Software Engineer | Scalable Systems | AI & Cloud Platforms

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SUMMARY

Final-year B.E. (AI & Data Science, 2026) software engineer with hands-on experience architecting, implementing, and testing scalable backend systems and AI-powered cloud platforms. Proven ability to design automation frameworks, write unit and functional tests, and use telemetry and metrics to drive operational excellence. Strong analytical and problem-solving skills; experienced collaborating cross-functionally with engineers and product stakeholders to ship high-quality, production-ready features.

WORK EXPERIENCE

AI Engineer Intern

Jan 2026 – May 2026

LearnersByte (AICTE Bharat Unnati AI Fellowship) | Remote

- Architected and delivered AI workflow modules end-to-end as engineering point of contact; managed sprint cycles, code reviews, and cross-functional reporting for a 5-engineer team over 4 months.
- Designed and implemented a scalable RAG pipeline; evaluated 6 chunking strategies and 3 embedding models, reducing average query latency from 800ms to under 300ms on a 50,000-document corpus with **22%** precision gain; tracked performance via latency, precision, and recall telemetry metrics.
- Built a transformer-based intent classifier on a 42-class dataset reaching **85%** accuracy; wrote unit and functional tests achieving **90%+** code coverage across pipeline components.

GenAI Data Science Analyst

Oct 2025 – Dec 2025

BCG X (Forage Virtual Internship) | Virtual

- Analyzed 3 financial datasets covering 5,000+ records; identified key risk indicators and delivered findings across 8 visualizations for a senior stakeholder review deck.
- Scoped an open-ended business problem into a structured AI solution statement; prototyped a scalable Q&A chatbot and presented 3 prioritized recommendations to simulated senior leadership within a 2-day deadline.

PROJECTS

Research Intelligence GraphRAG Platform – Enterprise Knowledge Retrieval

Python, Neo4j, LangChain, FAISS, BM25, Cross-Encoder, FastAPI, pytest, Docker

- Architected and delivered a scalable enterprise-grade knowledge platform over 100K+ scientific papers; constructed a Neo4j graph schema linking Papers, Authors, Datasets, Methods, and Citations to enable multi-hop reasoning at scale.
- Designed and implemented a high-performance retrieval pipeline combining semantic search, BM25, metadata filtering, graph traversal, and Reciprocal Rank Fusion (RRF) with cross-encoder re-ranking; instrumented telemetry across pipeline stages to track latency, throughput, and ranking quality metrics (Recall@K, MRR, NDCG).
- Developed automated test suites with pytest covering unit, functional, and integration scenarios; maintained **90%+** code coverage across retrieval, chunking, and re-ranking modules to ensure platform stability and release quality.

Multi-Agent Decision Intelligence Platform – Cloud-Native Agentic System

Python, FastAPI, LangGraph, OpenAI API, WebSockets, React, Docker, pytest

- Architected and delivered a cloud-native multi-agent platform with four specialized agents (Market Analyst, Financial Evaluator, Knowledge Retriever, Strategy Synthesizer); designed each agent as an isolated, independently testable sub-graph module to ensure platform stability and horizontal scalability.
- Managed inter-agent context via a structured shared state object with typed input/output keys, enabling deterministic handoffs and traceable reasoning; implemented automated integration test suites validating agent workflow correctness and achieving **85%+** code coverage across orchestration layers.
- Instrumented telemetry and latency metrics across agent execution to drive operational excellence; delivered real-time structured reports via WebSocket streaming, directly improving user-facing feature adoption and response quality.

SearchIQ – High-Performance Scalable Search Platform

Python, FastAPI, Sentence Transformers, FAISS, PostgreSQL, pytest, Docker

- Designed and implemented a high-performance, scalable semantic search platform using sentence embeddings and FAISS indexing; architected the backend for efficient retrieval across large document collections with sub-100ms query response under concurrent load.
- Built a test automation framework with pytest covering unit, functional, and performance test scenarios; drove code coverage to **88%+** and used query latency and throughput telemetry to identify bottlenecks and improve system reliability.
- Integrated LLM-based re-ranking and streamlined preprocessing pipelines, improving retrieval relevance over traditional keyword search while keeping the platform stable and scalable for production-grade usage.

SKILLS

Languages:	Python, Java, R, C
Software Engineering:	System Design, OOPs, Data Structures & Algorithms, Scalable Architecture, REST APIs
Testing & Quality:	pytest, Unit Testing, Functional Testing, Test Automation Frameworks, Code Coverage, Automated Test Suites
Observability:	Telemetry, Metrics-driven Development, Latency Monitoring, Operational Excellence
AI & ML:	LangChain, LangGraph, RAG Pipelines, LLM Orchestration, LlamaIndex, Scikit-learn, FAISS
Backend & APIs:	FastAPI, Flask, RESTful APIs, Redis Caching, Async Processing, WebSockets
Databases:	SQL, PostgreSQL, MongoDB, Neo4j
Cloud & DevOps:	Azure, AWS EC2, Docker, GitHub, Harness, CI/CD Pipelines

EDUCATION

B.E. in AI & Data Science	2023 – 2026
RNSIT Engineering College, Bengaluru	CGPA: 8.4 / 10 No Backlogs
🏆 Dr. R. N. Shetty Gold Medal	Branch Topper, AI & DS — VII Sem, VTU 2026

CERTIFICATIONS

Azure Data Fundamentals DP-900	Microsoft — Dec 2024
RAG for Enhanced AI Outputs	IBM SkillsBuild — Feb 2026
Intro to Large Language Models	IBM SkillsBuild — 2025
ChatGPT Prompt Engineering	DeepLearning.AI — Nov 2025
SQL Advanced Certification	HackerRank — Jun 2025
Data Science for Engineers	NPTEL, IIT Madras — Jan 2025

LEADERSHIP & ACHIEVEMENTS

Gold Medal — Branch Topper: Awarded the Dr. R. N. Shetty Medal of Merit by RNSIT for ranking first in AI & Data Science, VII Semester VTU Exams, Dec 2025/Jan 2026.

Technical Lead – LearnersByte: Led a 5-engineer team across 5 AI projects over 4 months; drove sprint planning, enforced code review standards, maintained test coverage, and delivered full project documentation on schedule.

Team Lead – College Project: Assigned modules, ran weekly standups, and presented a live system demo to a faculty and industry review panel.

Kaggle Contributor: Published ML/NLP notebooks with 200+ combined views; applied rigorous cross-validation and metric-focused evaluation practices.

LANGUAGES

English, Kannada, Hindi